ALPHA ARTS AND SCIENCE COLLEGE

(AFFILIATED TO UNIVERSITY OF MADRAS)
Porur, Chennai - 600 116

Record Note Book

Department of Computer Applications
(B.C.A) – Shift - I
Java Programming Practical (220C41)

Name : _____

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ALPHA ARTS AND SCIENCE COLLEGE
(AFFILIATED TO THE UNIVERSITY OF MADRAS)
Porur, Chennai –600 116

B.C.A PRACTICAL RECORD

Name: _____

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_____

Lecturer In-Charge **H.O.D**

Submitted for the B.C.A. Degree Practical Examination of **Java Programming Practical (220C41)** held on _____

Internal Examiner **External Examiner**

_____ _____

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Ex.No.1

PRIME NUMBER GENERATE UP TO GIVEN INTEGER

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.Scanner;

public class PrimeNumber {
public static void main(String arg[]) {
 int i, n, counter, j;

        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the n value : ");
        n = scanner.nextInt();

        System.out.println("Prime numbers between 1 to " + n + " are:");

        for (j = 2; j <= n; j++) {
            counter = 0;

            for (i = 1; i <= j; i++) {
                if (j % i == 0) {
                    counter++;
                }
            }

            if (counter == 2)
                System.out.print(j + " ");
        }
    }
}
```

OUTPUT:

```
Enter the n value : 10
Prime numbers between 1 to 10 are:
2 3 5 7
```

RESULT:

EX.NO.2

MULTIPLICATION OF TWO GIVEN MATRICES

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.*;
class matrixmul
{
public static void main(String args[])
{
Scanner s = new Scanner(System.in);
System.out.print("Enter number of rows for matrix1: ");
int row1 = s.nextInt();
System.out.print("Enter number of columns for matrix1: ");
int col1 = s.nextInt();
System.out.print("Enter number of rows for matrix2: ");
int row2 = s.nextInt();
System.out.print("Enter number of columns for matrix2: ");
int col2 = s.nextInt();
if (col1 != row2)
{
System.out.println("Matrix multiplication is not possible.");
return;
}
int a[][] = new int[row1][col1];
int b[][] = new int[row2][col2];
int c[][] = new int[row1][col2];
System.out.println("Enter elements of matrix1:");
for (int i = 0; i< row1; i++)
{
for (int j = 0; j < col1; j++)
{
a[i][j] = s.nextInt();
}
}
System.out.println("Enter elements of matrix2:");
for (int i = 0; i< row2; i++)
{
for (int j = 0; j < col2; j++)
{
b[i][j] = s.nextInt();
}
}
for (int i = 0; i< row1; i++)
{
```

```
        for (int j = 0; j < col2; j++)
            {
                c[i][j] = 0;
                for (int k = 0; k < col1; k++)
                {
                    c[i][j] =          c[i][j] + a[i][k] * b[k][j];
                }
            }
        }
System.out.println("Result of Matrix Multiplication:");
        for (int i = 0; i< row1; i++)
        {
            for (int j = 0; j < col2; j++)
            {
System.out.print(c[ i ][ j ]+ " ");
            }
System.out.println();
        }
    }
}
```

OUTPUT:

```
Enter number of rows for matrix1: 3
Enter number of columns for matrix1: 3
Enter number of rows for matrix2: 3
Enter number of columns for matrix2: 3
Enter elements of matrix1:
1 2 3
7 9 6
5 4 8
Enter elements of matrix2:
2 3 4
5 6 7
3 8 9
Result of Matrix Multiplication:
21 39 45
77 123 145
54 103 120
```

RESULT:

EX.NO:3

DISPLAY THE NUMBERS OF CHARACTERS,LINES AND WORDS IN A TEXT DOCUMENT

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.*;
class Count
{
   public static void main(String args[])
   {
      Scanner s = new Scanner(System.in);
      int lines = 0;
      int wordsc = 0;
      int chars = 0;
      int charsWithSpaces = 0;
      int charsWithoutSpaces = 0;
System.out.println("Enter the text (type 'exit' to stop): ");
      while (true)
      {
         String line = s.nextLine();

         if (line.equalsIgnoreCase("exit"))
         {
            break;
         }
lines++;
charsWithSpaces += line.length();
charsWithoutSpaces += line.replace(" ", "").length();
String words[] = line.split("\\s+");
wordsc += words.length;
      }

System.out.println("\nTotal Lines: " + lines);
System.out.println("Total Words: " + wordsc);
System.out.println("Total Characters with spaces: " + charsWithSpaces);
System.out.println("Total Characters without spaces: " + charsWithoutSpaces);
   }
}
```

OUTPUT:

```
Enter the text (type 'exit' to stop):
JAVA PROGRAMMING
PYTHON PROGRAMMING
HELLO WORLD
EXIT

Total Lines: 3
Total Words: 6
Total Characters with spaces: 45
Total Characters without spaces: 42
```

RESULT:

EX.NO.4

RANDOM NUMBERS GENERATION WITH A SPECIFIED RANGE

AIM:

ALG ORITHM:

PROGRAM:

```
import java.util.*;
public class Rand {
    public static void main(String[] args) {
    Scanner s=new Scanner(System.in);
 System.out.print("Enter the min value:");
    int min=s.nextInt();
 System.out.print("Enter the max limit:");
    int max=s.nextInt();
 System.out.print("How many Random numbers would you like to generate?:");
    int n=s.nextInt();
    for(int i=0;i<n;i++)
    {
     Random r = new Random();
  int num=r.nextInt(max-min)+min;
  System.out.println(num);
    }

          }
}
```

OUTPUT:

```
Enter the min value:3
Enter the max limit:10
How many Random numbers would you like to generate?:6
5
4
9
8
9
8
```

RESULT:

EX.NO.5

STRING MANIPULATION USING CHARACTER ARRAY

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.*;
class stringcal
{
  public static void main(String args[])
  {
    Scanner s=new Scanner(System.in);
System.out.println("Enter the string:");
    String str=s.nextLine();
    char len[]=str.toCharArray();
System.out.println("\nLength of the String:"+len.length);
System.out.print("\nEnter the position to find the character:");
    int position=s.nextInt();
    char x=len[position];
System.out.println("\ncharacter at position " +position+" is:"+x);
s.nextLine();
System.out.print("\nEnter the String to concatenate:");
    String s3=s.nextLine();
    String concat=str+s3;
System.out.println("\nconcatenated String:"+concat);
  }
}
```

OUTPUT:

```
Enter the string:
JAVA PROGRAMMING

Length of the String:17

Enter the position to find the character:3

character at position 3 is:A

Enter the String to concatenate: LANGUAGE

concatenated String:JAVA PROGRAMMING  LANGUAGE
```

RESULT:

EX.NO.6

STRING OPERATION USING THE STRING CLASS

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.*;

class stropns
{
   public static void main(String[] args)
   {
      Scanner s = new Scanner(System.in);

      System.out.print("Enter the first string: ");
      String string1 = s.nextLine();

      System.out.print("Enter the second string: ");
      String string2 = s.nextLine();

      int ch;

      do {
         System.out.println("\n1. Concatenate");
         System.out.println("2. Search");
         System.out.println("3. Extract");
         System.out.println("4. Exit");
         System.out.print("Enter the choice: ");
         ch = s.nextInt();
         s.nextLine();   // clear buffer

         switch (ch)
         {
            case 1:
               String concat = string1 + " " + string2;
               System.out.println("The concatenated string is: " + concat);
               break;

            case 2:
               System.out.print("Enter the string for searching in first string: ");
               String s1 = s.nextLine();

               if (string1.contains(s1))
                  System.out.println("Substring found at index: " + string1.indexOf(s1));
               else
                  System.out.println("Substring not found.");
               break;



            case 3:
```

```
            System.out.print("Enter the substring to extract from second string: ");
            String search2 = s.nextLine();

            if (string2.contains(search2))
            {
                int startIndex = string2.indexOf(search2);
                int endIndex = startIndex + search2.length();
                String finalstring = string2.substring(startIndex, endIndex);
                System.out.println("Extracted substring is: " + finalstring);
            }
            else
            {
                System.out.println("The given substring is not found in second string.");
            }
            break;

        case 4:
            System.out.println("End.");
            break;

        default:
            System.out.println("Invalid choice");
        }

    } while (ch != 4);

    s.close();
  }
}
```

OUTPUT:

```
Enter the first string: PROGRAMMING
Enter the second string: LANGUAGE

1. Concatenate
2. Search
3. Extract
4. Exit
Enter the choice: 1
The concatenated string is: PROGRAMMING LANGUAGE

1. Concatenate
2. Search
3. Extract
4. Exit
Enter the choice: 2
Enter the string for searching in first string: PROGRAMMING
Substring found at index: 0

1. Concatenate
2. Search
3. Extract
4. Exit
Enter the choice: 3
Enter the substring to extract from second string: LANGUAGE
Extracted substring is: LANGUAGE

1. Concatenate
2. Search
3. Extract
4. Exit
Enter the choice: 4
End.
```

RESULT:

EX.NO.7

STRING OPERATION USING THE STRINGBUFFER CLASS

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.*;
public class stropnstringbuffer {
   public static void main(String[] args) {
      Scanner s=new Scanner(System.in);
      System.out.print("Enter a String1:");
      String str1=s.nextLine();
      System.out.print("Enter a String2:");
      String str2=s.nextLine();
      StringBuffer app1 = new StringBuffer(str1);
      StringBuffer app2=new StringBuffer(str2);
      int choice;
            do {
         System.out.println("Main Menu:");
         System.out.println("1. Append");
         System.out.println("2. Insert");
         System.out.println("3. Replace");
         System.out.println("4. Delete");
         System.out.println("5. Reverse");
         System.out.println("6. Exit");
         System.out.println("Enter your choice: ");
         choice = s.nextInt();
         s.nextLine();
                  switch (choice) {
            case 1:
      System.out.println("Append.");
      StringBuffer app = app1.append(app2);
      System.out.println("After append: " + app);
               break;
         case 2:
      System.out.println("Insert.");
      System.out.println("Current String: " + app1);
        System.out.print("Enter a String to insert:");
      String str3=s.nextLine();
   System.out.println("Enter the position to insert:");
     int pos=s.nextInt();
    StringBuffer app3=new StringBuffer(str3);
     app1.insert(pos, app3);
     System.out.println("After insert: " + app1);
         break;
```

```
        case 3:
         s.nextLine();
          System.out.print("Enter a String to replace:");
        String str4=s.nextLine();
    System.out.println("Enter the starting index1:");
      int si1=s.nextInt();
    System.out.println("Enter the ending index2:");
      int si2=s.nextInt();
            app1.replace(si1, si2, str4);
        System.out.println("After replace: " + app1);
                break;
            case 4:
            System.out.println("Enter the starting index1:");
      int s1=s.nextInt();
    System.out.println("Enter the ending index2:");
      int s2=s.nextInt();
                        app1.delete(s1, s2);
          System.out.println("After delete: " + app1);
          break;
         case 5:
        app1.reverse();
        System.out.println("After reverse: " + app1);
        break;
            case 6:
        System.out.println("Exiting the program. Goodbye!");
                break;
              default:
System.out.println("Invalid choice. Please try again.");
                break;
            }
            System.out.println();
        } while (choice != 6);
    }
}
```

OUTPUT:

```
Enter a String1:HELLO WORLD
Enter a String2:JAVA PROGRAMMING
Main Menu:
1. Append
2. Insert
3. Replace
4. Delete
5. Reverse
6. Exit
Enter your choice:
1
Append.
After append: HELLO WORLDJAVA PROGRAMMING

Main Menu:
1. Append
2. Insert
3. Replace
4. Delete
5. Reverse
6. Exit
Enter your choice:
2
Insert.
Current String: HELLO WORLDJAVA PROGRAMMING
Enter a String to insert:PYTHON
Enter the position to insert:
6
After insert: HELLO PYTHONWORLDJAVA PROGRAMMING
```

```
Main Menu:
1. Append
2. Insert
3. Replace
4. Delete
5. Reverse
6. Exit
Enter your choice:
3

Enter a String to replace:PROGRAMMING
Enter the starting index1:
12
Enter the ending index2:
17
After replace: HELLO PYTHONPROGRAMMINGJAVA PROGRAMMING

Main Menu:
1. Append
2. Insert
3. Replace
4. Delete
5. Reverse
6. Exit
Enter your choice:
4
Enter the starting index1:
0
Enter the ending index2:
5
After delete:  PYTHONPROGRAMMINGJAVA PROGRAMMING
```

```
Main Menu:
1. Append
2. Insert
3. Replace
4. Delete
5. Reverse
6. Exit
Enter your choice:
5
After reverse: GNIMMARGORP AVAJGNIMMARGORPNOHTYP

Main Menu:
1. Append
2. Insert
3. Replace
4. Delete
5. Reverse
6. Exit
Enter your choice:
6
Exiting the program. Goodbye!
```

RESULT:

EX.NO.8

MULTI-THREAD APPLICATION FOR RANDOM NUMBER GENERATION AND SQUARE & CUBE CALCULATION

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.Random;
public class EXNO8 {
 public static void main(String args[])
 {
 Number n = new Number();
 n.start();
 }
}
class Square extends Thread
{
int x;
Square(int n)
{
x = n;
}
public void run()
{
int sqr = x * x;
System.out.println("Square of " + x + " = " + sqr );
}
}
class Cube extends Thread
{
int x;
Cube(int n)
{
x = n;
}
public void run()
{
int cub = x * x * x;
System.out.println("Cube of " + x + " = " + cub );
}
```

```
class Number extends Thread
{
public void run()
{
Random random = new Random();
for(int i =0; i<10; i++)
{
int randomInteger = random.nextInt(100);
System.out.println("Random Integer generated : " + randomInteger);
Square s = new Square(randomInteger);
s.start();
Cube c = new Cube(randomInteger);
c.start();
try {
Thread.sleep(1000);
}
catch (InterruptedException ex) {
System.out.println(ex);
                }
            }
        }
    }
```

OUTPUT:

```
Random Integer generated : 52
Cube of 52 = 140608
Square of 52 = 2704
Random Integer generated : 50
Cube of 50 = 125000
Square of 50 = 2500
Random Integer generated : 91
Square of 91 = 8281
Cube of 91 = 753571
Random Integer generated : 22
Square of 22 = 484
Cube of 22 = 10648
Random Integer generated : 1
Square of 1 = 1
Cube of 1 = 1
Random Integer generated : 82
Square of 82 = 6724
Cube of 82 = 551368
Random Integer generated : 2
Cube of 2 = 8
Square of 2 = 4
Random Integer generated : 39
Square of 39 = 1521
Cube of 39 = 59319
Random Integer generated : 8
Square of 8 = 64
Cube of 8 = 512
Random Integer generated : 50
Cube of 50 = 125000
Square of 50 = 2500
```

EX.NO.9

MULTI-THREADING TO PRINT THE SPECIFIED NUMBERS

AIM:

ALGORITHM:

PROGRAM:

```
public class ThreadNumberPrinter
{
   public static void main(String[]args)
   {
      NumPrint n1=new NumPrint(1,10);
      NumPrint n2=new NumPrint(40,50);
      n1.start();
      n2.start();
   }
}
class NumPrint extends Thread
{
   int start,end;
   NumPrint(int x,int y)
   {
      start=x;
      end=y;
   }
   public void run()
   {
      for(int i=start;i<=end;i++)
      {
         System.out.println(i);
         try
         {
            Thread.sleep(1000);
         }
         catch(Exception e)
         {
         }
      }
   }
}
```

OUTPUT:

```
1
40
2
41
3
42
4
43
5
44
6
45
7
46
8
47
9
48
49
10
50
```

RESULT:

EX.NO.10

EXCEPTION HANDLING USING PRE-DEFINED EXCEPTION

AIM:

ALGORITHM:

PROGRAM:

```
import java.util.*;

class EXNO10
{
   public static void main(String args[])
   {
      Scanner s = new Scanner(System.in);

      int n1, n2, n3, d, size, ch;
      int arr1[] = {25, 30, 15};
      int arr2[];

         System.out.println("EX.NO.10: EXCEPTION HANDLING USING PREDEFINED
EXCEPTIONS");

      System.out.println("1. Arithmetic Exception");
      System.out.println("2. Number Format Exception");
      System.out.println("3. Array Index Out Of Bounds Exception");
      System.out.println("4. Negative Array Size Exception");
      System.out.println("5. Exit");

      do
      {
         System.out.println("\nEnter your choice:");
         ch = s.nextInt();

         try
         {
            switch(ch)
            {
               case 1:
                  n1 = 4;
                  n2 = 0;
                  n3 = n1 / n2;
                  System.out.println("The result of division is = " + n3);
                  break;

               case 2:
                  d = Integer.parseInt("12x");
                  System.out.println(d);
                  break;
```

```
      case 3:
          System.out.println("The Elements in the array are:");
          for(int i = 1; i <= 3; i++)
             System.out.println(arr1[i]);
          break;

        case 4:
          System.out.println("Enter the array size to create an array:");
          size = s.nextInt();
          arr2 = new int[size];
          System.out.println("The array created successfully");
          break;

        case 5:
          System.out.println("Exiting the program...");
          return;

        default:
          System.out.println("Invalid choice. Enter a number between 1 and 5");
    }
}

catch(ArithmeticException ae)
{
    System.out.println("Error: " + ae.getMessage());
    System.out.println("Division by Zero occurred. Performing valid division");

    n1 = 4;
    n2 = 2;
    n3 = n1 / n2;

    System.out.println("The Result of division is = " + n3);
}

catch(NumberFormatException nfe)
{
    System.out.println("Error: " + nfe.getMessage());
    System.out.println("Invalid input detected. Correcting by parsing valid input");

    d = Integer.parseInt("12");
    System.out.println("The Correct Number is = " + d);
}
```

```
        catch(ArrayIndexOutOfBoundsException aoe)
          {
             System.out.println("Error: " + aoe.getMessage());
             System.out.println("Printing valid elements of the array:");

             for(int i = 0; i < 3; i++)
                System.out.println(arr1[i]);
          }

          catch(NegativeArraySizeException nase)
          {
                   System.out.println("Negative array size is not allowed. Please enter a positive
size:");
             size = s.nextInt();
             arr2 = new int[size];
             System.out.println("The array created successfully");
          }

       } while(true);
    }
}
```

OUTPUT:

```
EX.NO.10: EXCEPTION HANDLING USING PREDEFINED EXCEPTIONS
1. Arithmetic Exception
2. Number Format Exception
3. Array Index Out Of Bounds Exception
4. Negative Array Size Exception
5. Exit

Enter your choice:
1
Error: / by zero
Division by Zero occurred. Performing valid division
The Result of division is = 2

Enter your choice:
2
Error: For input string: "12x"
Invalid input detected. Correcting by parsing valid input
The Correct Number is = 12

Enter your choice:
3
The Elements in the array are:
30
15
Error: Index 3 out of bounds for length 3
Printing valid elements of the array:
25
30
15
```

```
Enter your choice:
4
Enter the array size to create an array:
-5
Negative array size is not allowed. Please enter a positive size:
3
The array created successfully

Enter your choice:
4
Enter the array size to create an array:
-5
Negative array size is not allowed. Please enter a positive size:
3
The array created successfully

Enter your choice:
5
Exiting the program...
```

RESULT:

EX.NO.11

DISPLAY FILE INFORMATION AND PROPERTIES

AIM:

ALOGORITHM:

PROGRAM:

```
import java.io.*;
import java.util.*;
class Main
{
public static void main(String[] args)
{
System.out.print("Enter the file name : ");
Scanner s = new Scanner(System.in);
String filename = s.nextLine();
File fobj = new File(filename);
if (fobj.exists())
{
System.out.println("Details about File");
System.out.println("File exists: Yes");
System.out.println("Readable: " + (fobj.canRead() ? "Yes" : "No"));
System.out.println("Writable: " + (fobj.canWrite() ? "Yes" : "No"));
String fileType = " ";
String fileExten = fobj.getName().toLowerCase();
if (fileExten.endsWith(".doc"))
{
fileType = "Word File";
}
else if (fileExten.endsWith(".java"))
{ fileType = "Java Source File";
}
else if (fileExten.endsWith(".class"))
{ fileType = "Java Compiled Class File";
}
else if (fileExten.endsWith(".jpg") || fileExten.endsWith(".jpeg"))
{ fileType = "JPEG Image File";
}
else if (fileExten.endsWith(".pdf"))
{ fileType = "PDF Document";
}
else
{ fileType = " File type other than Word, Java, image and pdf";
}
System.out.println("File Type: " + fileType);
System.out.println("File Length: " + fobj.length() + " bytes\n ");
}
else
{ System.out.println("The file does not exist.\n");
} } }
```

OUTPUT:

```
Enter the file name : Main.java
Details about File
File exists: Yes
Readable: Yes
Writable: Yes
File Type: Java Source File
File Length: 1268 bytes
```

RESULT:

EX.NO.12

TEXT FORMATTING USING JAVA AWT

AIM:

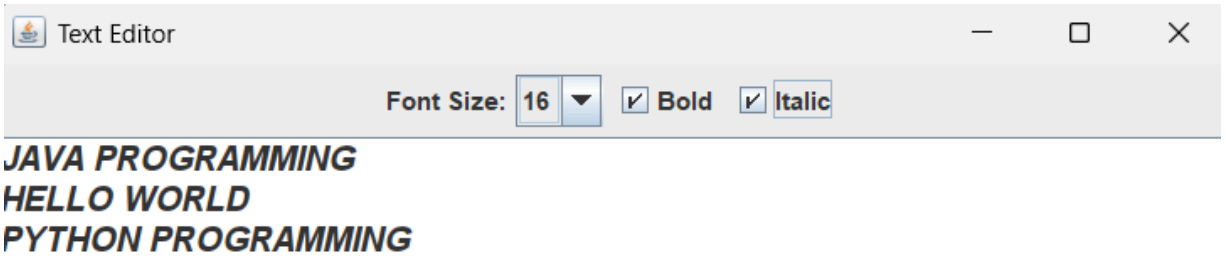
ALGORITHM:

PROGRAM:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class TextEditorss extends JFrame implements ActionListener {

    private JTextArea textArea;
    private JComboBox<String> fontSizeCombo;
    private JCheckBox boldCheckBox;
    private JCheckBox italicCheckBox;

    public TextEditorss() {

        setTitle("Text Editor");
        setSize(600, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout());

        textArea = new JTextArea();
        textArea.setFont(new Font("Arial", Font.PLAIN, 14));

        JScrollPane scrollPane = new JScrollPane(textArea);
        add(scrollPane, BorderLayout.CENTER);

        JPanel controlPanel = new JPanel();
        controlPanel.setLayout(new FlowLayout());

        fontSizeCombo = new JComboBox<>();
        fontSizeCombo.addItem("12");
        fontSizeCombo.addItem("14");
        fontSizeCombo.addItem("16");
        fontSizeCombo.addItem("18");
        fontSizeCombo.addItem("20");
        fontSizeCombo.setSelectedIndex(1);
        fontSizeCombo.addActionListener(this);

        controlPanel.add(new JLabel("Font Size:"));
        controlPanel.add(fontSizeCombo);

        boldCheckBox = new JCheckBox("Bold");
        boldCheckBox.addActionListener(this);
        controlPanel.add(boldCheckBox);
```

```
        italicCheckBox = new JCheckBox("Italic");
        italicCheckBox.addActionListener(this);
        controlPanel.add(italicCheckBox);

        add(controlPanel, BorderLayout.NORTH);

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {

        int fontSize = Integer.parseInt((String) fontSizeCombo.getSelectedItem());

        int fontStyle = Font.PLAIN;

        if (boldCheckBox.isSelected()) {
            fontStyle = fontStyle | Font.BOLD;
        }

        if (italicCheckBox.isSelected()) {
            fontStyle = fontStyle | Font.ITALIC;
        }

        Font currentFont = textArea.getFont();
        Font newFont = new Font(currentFont.getFontName(), fontStyle, fontSize);

        textArea.setFont(newFont);
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> new TextEditorss());
    }
}
```

OUTPUT:


Text Editor
Font Size:
16
Bold
Italic
JAVA PROGRAMMING
HELLO WORLD
PYTHON PROGRAMMING

RESULT:

EX.NO.13

MOUSE EVENTS HANDLING USING ADAPTER CLASSES

AIM:

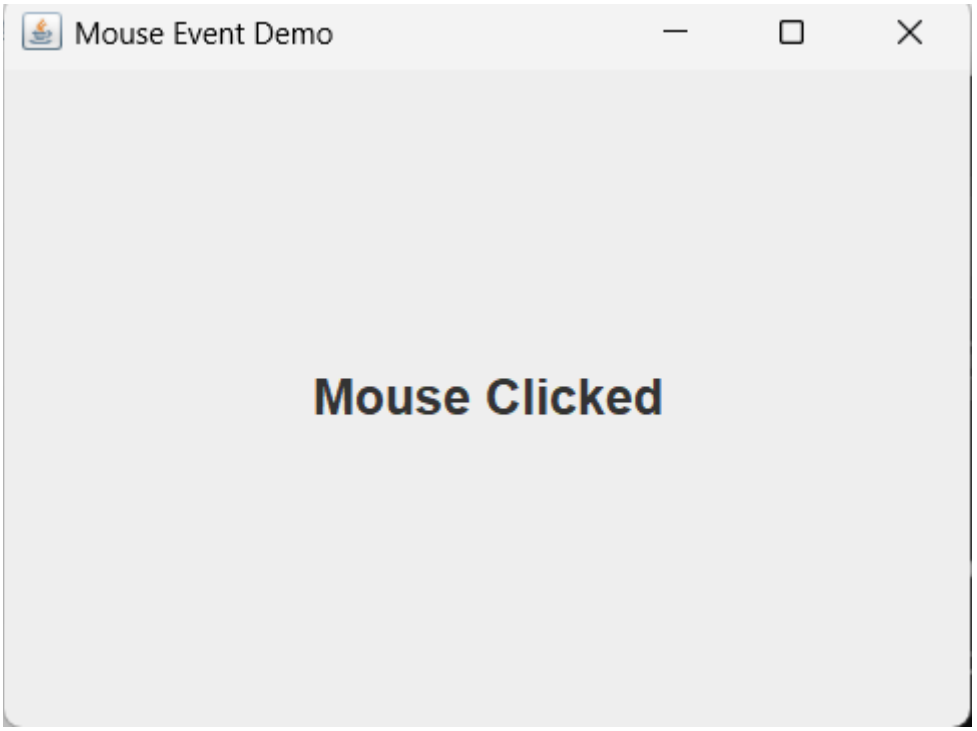
ALOGORITHM:

PROGRAM:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class MouseEventDemo extends JFrame {
    private JLabel eventLabel;

    public MouseEventDemo() {
        setTitle("Mouse Event Demo");
        setSize(400, 300);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        eventLabel = new JLabel("Move mouse over the window", JLabel.CENTER);
        eventLabel.setFont(new Font("Arial", Font.BOLD, 20));
        add(eventLabel, BorderLayout.CENTER);

        addMouseListener(new MouseAdapter() {
            public void mouseClicked(MouseEvent e) {
                displayEventName("Mouse Clicked");
            }

            public void mousePressed(MouseEvent e) {
                displayEventName("Mouse Pressed");
            }

            public void mouseReleased(MouseEvent e) {
                displayEventName("Mouse Released");
            }

            public void mouseEntered(MouseEvent e) {
                displayEventName("Mouse Entered");
            }

            public void mouseExited(MouseEvent e) {
                displayEventName("Mouse Exited");
            }
        });

        setVisible(true);
    }
```

```
private void displayEventName(String eventName) {
eventLabel.setText(eventName);
}
public static void main(String[] args) {
SwingUtilities.invokeLater(() -> new MouseEventDemo());
}
}
```

OUTPUT:


Mouse Event Demo
Mouse Clicked

RESULT:

EX.NO.14

SIMPLE CALCULATOR USING JAVA AWT

AIM:

ALGORITHM:

PROGRAM:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Calculator extends JFrame implements ActionListener {
    private JTextField displayField;

    public Calculator() {
        setTitle("Simple Calculator");
        setSize(300, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        displayField = new JTextField();
        displayField.setEditable(false); // Make it read-only
        displayField.setHorizontalAlignment(JTextField.RIGHT);
        add(displayField, BorderLayout.NORTH);

        JPanel buttonPanel = new JPanel(new GridLayout(4, 4, 5, 5));

        addButton(buttonPanel, "7");
        addButton(buttonPanel, "8");
        addButton(buttonPanel, "9");
        addButton(buttonPanel, "/");

        addButton(buttonPanel, "4");
        addButton(buttonPanel, "5");
        addButton(buttonPanel, "6");
        addButton(buttonPanel, "*");

        addButton(buttonPanel, "1");
        addButton(buttonPanel, "2");
        addButton(buttonPanel, "3");
        addButton(buttonPanel, "-");

       addButton(buttonPanel, "0");
        addButton(buttonPanel, ".");
        addButton(buttonPanel, "=");
        addButton(buttonPanel, "+");




 add(buttonPanel, BorderLayout.CENTER);
```

```
        setVisible(true);
    }

    private void addButton(Container container, String text) {
        JButton button = new JButton(text);
        button.addActionListener(this);
        container.add(button);
    }

    public void actionPerformed(ActionEvent e) {
        String command = e.getActionCommand();

        if ("0123456789.".contains(command)) {
            displayField.setText(displayField.getText() + command);
        } else if (command.equals("+") || command.equals("-") || command.equals("*") ||
command.equals("/")) {
            displayField.setText(displayField.getText() + " " + command + " ");
        } else if (command.equals("=")) {
            evaluateExpression();
        }
    }

    private void evaluateExpression() {
        String expression = displayField.getText();
        String[] parts = expression.split(" ");

      if (parts.length != 3) {
            displayField.setText("Invalid expression");
            return;
        }

        try {
            double operand1 = Double.parseDouble(parts[0]);
            String operator = parts[1];
            double operand2 = Double.parseDouble(parts[2]);
            double result = 0;





        switch (operator) {
                case "+":
```

```
                result = operand1 + operand2;
                break;
            case "-":
                result = operand1 - operand2;
                break;
            case "*":
                result = operand1 * operand2;
                break;


            case "/":
                if (operand2 == 0) {
                    displayField.setText("Error: Divide by zero");
                    return;
                }
                result = operand1 / operand2;
                break;
            default:
                displayField.setText("Invalid operator");
                return;
        }

        displayField.setText(Double.toString(result));
    } catch (NumberFormatException ex) {
        displayField.setText("Invalid expression");
    }
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new Calculator());
}
}
```

OUTPUT:

Simple Calculator
78 + 95
7
8
9
/
4
5
6
*
1
2
3
-
0
.
=
+

Simple Calculator
173.0
7
8
9
/
4
5
6
*
1
2
3
-
0
.
=
+

RESULT:

EX.NO.15

TRAFFIC LIGHT SIMULATOR USING RADIO BUTTONS

AIM:

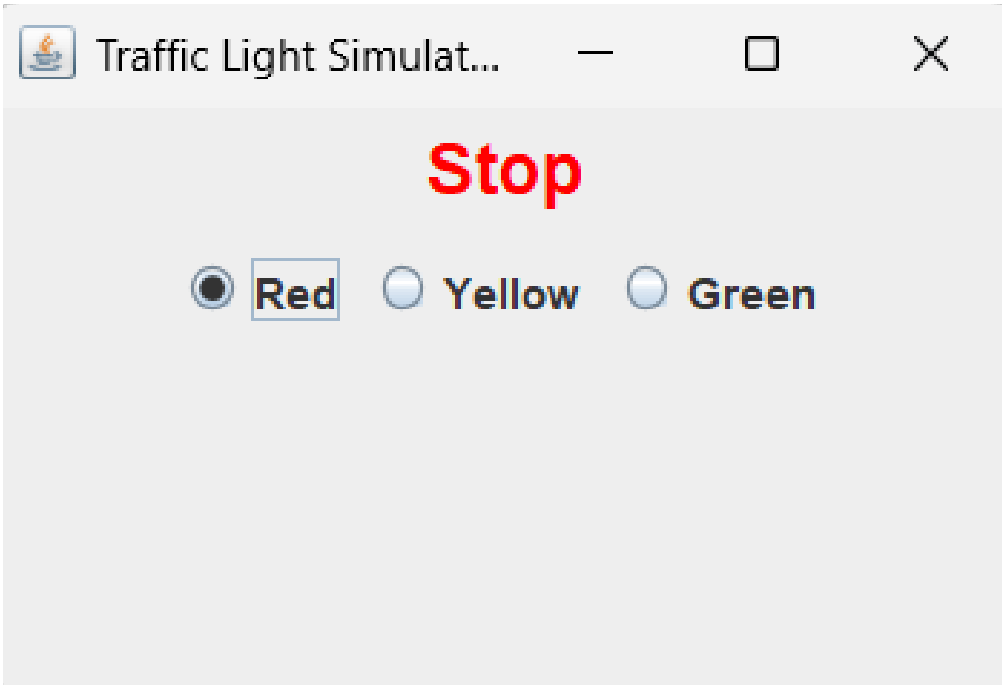
ALGORITHM:

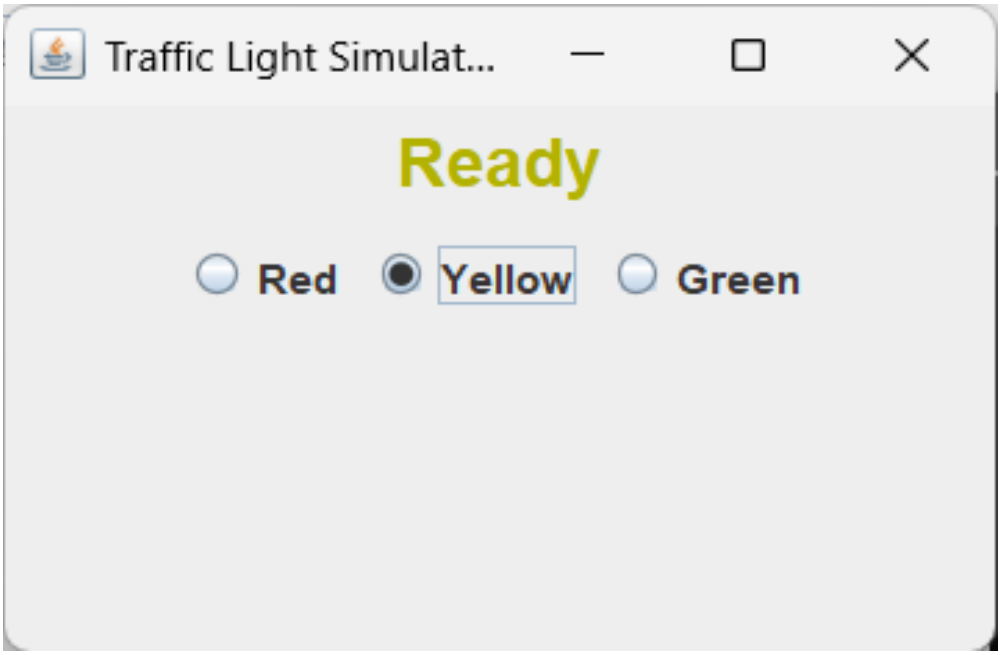
PROGRAM:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class TrafficLightSimulator extends JFrame implements ActionListener {
    private JLabel messageLabel;
    private JRadioButton redButton, yellowButton, greenButton;

    public TrafficLightSimulator() {
        setTitle("Traffic Light Simulator");
        setSize(300, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout());

        JPanel radioPanel = new JPanel(new FlowLayout());

        redButton = new JRadioButton("Red");
        yellowButton = new JRadioButton("Yellow");
        greenButton = new JRadioButton("Green");

        ButtonGroup buttonGroup = new ButtonGroup();
        buttonGroup.add(redButton);
        buttonGroup.add(yellowButton);
        buttonGroup.add(greenButton);

        redButton.addActionListener(this);
        yellowButton.addActionListener(this);
        greenButton.addActionListener(this);

        radioPanel.add(redButton);
        radioPanel.add(yellowButton);
        radioPanel.add(greenButton);

        JPanel messagePanel = new JPanel();
        messagePanel.setLayout(new FlowLayout());
        messageLabel = new JLabel("No message", JLabel.CENTER);
        messageLabel.setFont(new Font("Arial", Font.BOLD, 20));
        messagePanel.add(messageLabel);





        add(radioPanel, BorderLayout.CENTER);
        add(messagePanel, BorderLayout.NORTH);
```

```
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == redButton) {
            messageLabel.setText("Stop");
            messageLabel.setForeground(Color.RED);
        } else if (e.getSource() == yellowButton) {
            messageLabel.setText("Ready");
            messageLabel.setForeground(Color.YELLOW.darker()); // Darker yellow
        } else if (e.getSource() == greenButton) {
            messageLabel.setText("Go");
            messageLabel.setForeground(Color.GREEN.darker()); // Darker green
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> new TrafficLightSimulator());
    }
}
```

OUTPUT:


Traffic Light Simulat...
Stop
Red
Yellow
Green


Traffic Light Simulat...
Ready
Red
Yellow
Green

Traffic Light Simulat...
Go
Red
Yellow
Green

RESULT: